Yan Xie, PhD

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♦ Research interests

Clouds and precipitation, Trustworthy AI and machine learning, Radar and satellite remote sensing

♦ Education

University of Michigan, Ann Arbor *Aug 2019 – May 2025*
Ph.D., Climate and Space Sciences and Engineering (CLaSP) (GPA: 4.0 / 4.0)
Advisors: Prof. Claire Pettersen, Prof. Xianglei Huang
Graduate coursework: Machine Learning (EECS 545), Bayesian Data Analysis (STATS 451), Advanced Data Analysis, Advanced Fluid Dynamics, Climate Physics, Radiative Transfer, Remote Sensing

McGill University *Jul – Oct 2018*
Undergraduate Research Trainee

Nanjing University *Sep 2015 – Jun 2019*
B.Sc., Atmospheric Science, Graduate with honors (GPA: 3.8 / 4.0)

♦ Work experience

University of Oklahoma, School of Meteorology **–** Postdoctoral Research Associate *Jul 2025 – present*
University of Michigan, Dept. of CLaSP – Research Fellow *May – Jun 2025*
University of Michigan, Dept. of CLaSP – Graduate Student Research Assistant *Aug 2019 – Jun 2025*

♦ Publications

Peer-reviewed

1. King, F., Pettersen, C., Posselt, D., Ringerud, S., & **Xie, Y.** (2025). Leveraging Sparse Autoencoders to Reveal Interpretable Features in Geophysical Models. *Journal of Geophysical Research: Machine Learning and Computation*, 3, doi:10.1029/2025JH000769
2. **Xie, Y.**, King, F., Pettersen, C., & Flanner, M. (2025). Machine Learning detection of melting layers from radar observations. *Journal of Geophysical Research: Machine Learning and Computation*, 3, doi:10.1029/2024JH000521
3. **Xie, Y.**, Pettersen, C., Flanner, M., & Shates, J. (2024). Ground-observed snow albedo changes during rain-on-snow events in northern Alaska. *Journal of Geophysical Research: Atmospheres*, 129, doi:10.1029/2024JD040975
4. Liu, Y., Huang, Y., Yuan, J., **Xie, Y.**, & Zhou, C. (2024). Contribution of surface radiative effects, heat fluxes and their interactions to land surface temperature variability. *Journal of Geophysical Research: Atmospheres*, 129, doi:10.1029/2023JD039495
5. **Xie, Y.**, Huang, X., Chen, X., L'Ecuyer, T. S., & Drouin, B. J. (2023). Joint use of far-infrared and mid-infrared observation for sounding retrievals: Learning from the past for upcoming far-infrared missions. *Earth and Space Science*, 10, doi:10.1029/2022EA002684
6. **Xie, Y.**, Huang, X., Chen, X., L'Ecuyer, T. S., Drouin, B. J., & Wang, J. (2022). Retrieval of Surface Spectral Emissivity in Polar Regions Based on the Optimal Estimation Method. *Journal of Geophysical Research: Atmospheres*, 127, doi:10.1029/2021JD035677
7. L'Ecuyer, T. S., Drouin, B. J., Anheuser, J., Grames, M., Henderson, D. S., Huang, X., Kahn, B. H., Kay, J. E., Lim, B. H., Mateling, M., Merrelli, A., Miller, N. B., Padmanabhan, S., Peterson, C., Schlegel, N., White, M. L., & **Xie, Y.** (2021). The Polar Radiant Energy in the Far Infrared Experiment: A New Perspective on Polar Longwave Energy Exchanges. *Bulletin of the American Meteorological Society*, *102*(7), doi:10.1175/bams-d-20-0155.1
8. Huang, Y., Chou, G., **Xie, Y.**, & Soulard, N. (2019). Radiative Control of the Interannual Variability of Arctic Sea Ice. *Geophysical Research Letters*, 46, 9899–9908, doi:10.1029/2019gl084204

Submitted and in revision

1. **Xie, Y.**, Pettersen, C., & King, F. Long-term observations reveal more frequent and persistent Arctic stratiform rainfall. *Communications Earth & Environment* [submitted]
2. **Xie, Y.,** McGovern, A., Bostrom, A., & Rao, D. Improving AI forecasts via bias correction: applications on heatwaves. *Artificial Intelligence for the Earth Systems* [submitted]
3. Erickson, N., Huynh, N., **Xie, Y.**, McGovern, A., & Bostrom, A. On global data representativeness in artificial intelligence for meteorology. *Bulletin of the American Meteorological Society* [in revision]

In preparation

1. **Xie, Y.**, McGovern, A., Bostrom, A., Rao, D., Loveday, N., & Kaandorp, M. Bias detection and correction for Artificial Intelligence-based weather prediction. Target: *Artificial Intelligence for the Earth Systems*

♦ Research Experience

Bias detection and correction for improving AI weather prediction *Jul 2025 – Present*
Postdoctoral Research Associate *School of Meteorology, University of Oklahoma*
Mentors: Prof. Amy McGovern, Prof. Ann Bostrom

- Implemented and trained the NSF NCAR CREDIT AI weather prediction framework on 2000-2025 ERA5 reanalysis at 0.25° resolution, using multi-GPU HPC nodes (4× NVIDIA A100) with distributed training and hyperparameter optimization; ran pretrained AI weather models (ECMWF AIFS, GraphCast, FuXi) through NVIDIA Earth2Studio
- Evaluated the impacts of bias correction on AI-based forecasts of extreme weather events, reduced heatwave-day surface temperature biases in AI forecasts by 50-90% across global land, and revealed geographic disparities in AI forecast improvements resulting from bias correction (*AIES*, submitted)
- Contributed to assessing global data representativeness in AI weather prediction (*BAMS*, in revision)
- Leading a review assessing bias correction methods across weather and climate models, including their applicability and limitations for AI-powered predictive models (*AIES*, in prep)

Arctic stratiform clouds and rainfall using machine learning *Oct 2022 – Jun 2025*
Graduate Student Research Assistant *Dept. of CLaSP, University of Michigan*
Mentors: Prof. Claire Pettersen, Prof. Mark Flanner, Dr. Fraser King

- Developed a U-Net model to detect precipitation melting layers (rain-snow phase transitions) in DOE ARM cloud and precipitation radar observations at the North Slope of Alaska sites, with Monte Carlo dropout uncertainty estimates and interpretability analyses (*JGR: MLC*, 2025)
- Applied the U-Net model to 28 years (1998-2025) of DOE ARM cloud radar observations, and revealed more frequent and persistent Arctic stratiform rainfall (*Commun. Earth Environ.*, submitted)
- Quantified rain-on-snow impacts on surface snow albedo, ran the SNICAR-ADv3 in the CESM land model and assessed simulations against ground observations (*JGR: Atmospheres*, 2024)
- Contributed to interpretable AI for geophysical models using sparse autoencoders (*JGR: MLC, 2025*)

Satellite retrievals of polar atmospheric profiles and surface properties *Sep 2019 – Sep 2022*
Graduate Student Research Assistant *Dept. of CLaSP, University of Michigan*
Mentors: Prof. Xianglei Huang, Dr. Xiuhong Chen

- Developed an optimal-estimation based Bayesian framework to retrieve infrared surface spectral emissivity using satellite observations over polar regions (*JGR: Atmospheres*, 2022; *ESS*, 2023)
- Demonstrated the influence of water vapor absorption and spectral resolution on infrared sounding retrievals, informing NASA's PREFIRE polar far-infrared mission (*BAMS*, 2021)

♦ Selected Presentations

1. **Xie, Y.**, Pettersen, C., & King, F. "More frequent and persistent stratiform rainfall at a long-term Arctic observatory". AMS Madison Summit. 2026 (Oral)
2. **Xie, Y.**, McGovern, A., Bostrom, A., & Rao, D. "Bias correction in data preparation and its implications for AI models". AMS 106th Annual Meeting. 2026 (Oral)
3. **Xie, Y.**, King, F., Pettersen, C., & Flanner, M. "Melting Layer Detection from Radar Observations Using Machine Learning". AGU 2024 Fall Meeting. (Oral; **Outstanding Student Presentation Award**)
4. **Xie, Y.**, Pettersen, C., Flanner, M., & Shates, J. "Ground-observed Influence of Rainfall on Surface Snow Albedo at North Slope of Alaska". AGU 2023 Fall Meeting. (eLightning)

5. **Xie, Y.**, Huang, X., Chen, X., L'Ecuyer, T. S., & Drouin, B. J. "On the use of far-IR radiances in satellite retrievals: how can the observations collected half century ago help us preparing for the upcoming missions". AGU 2022 Fall Meeting. (Oral)
6. **Xie, Y.**, Huang, X., & Chen., X. "Retrieval of surface spectral emissivity in the polar regions: an optimal-estimation approach". AGU 2020 Fall Meeting. (Oral)

♦ Teaching and mentoring

Graduate Student Instructor, CLIMATE 105 *Our Changing Climate*, University of Michigan *Winter 2023*
- Monitored class discussions for 80+ undergraduates, held weekly office hours, graded assignments

Graduate Student Peer Mentor, University of Michigan *Fall 2023*
- Mentored a graduate student, provided guidance and feedback on academics and student life

Research mentoring, University of Oklahoma *Jan 2026 – present*
- Mentored a graduate student, provided feedback on research methodology and manuscript, leading to a paper currently under revision at *BAMS*

Guest lectures and outreach *Feb 2026 – present*
- Invited guest lecturer at the Midwest Climate Adaptation Science Center hosted by the University of Minnesota, and a mentor for K-12 students in the Open Ideas Lab's "Save Mumbai 2050" initiative

♦ Professional service

Associate Editor for *Artificial Intelligence for the Earth Systems* *Sep 2026 – present*
Panelist for NASA's 2025 ROSES Earth Venture research proposal review panel *Mar 2026*
Reviewer for journals including *JGR: Atmospheres, Artificial Intelligence for the Earth Systems, Journal of Hydrology, Water Resources Research, Atmospheric Chemistry and Physics*, etc. *2025 – present*
Member, AGU Precipitation Technical Committee *2024 – present*
Organizing Committee, Michigan Geophysical Union *Jan – Apr 2023*

♦ Honors and awards

AGU Outstanding Student Presentation Award (top 3-5%) *2025*
AGU Precipitation Technical Committee Student Award *2024*
Rackham Predoctoral Fellowship, University of Michigan *2024*
Winning team project, NSF I-GUIDE Summer School: "Assessing the economic impact of wildfire on building losses in Texas" *2024*

♦ Technical skills

AI and machine learning: PyTorch, TensorFlow, development and training of U-Net and Transformer-based models, AI weather prediction (NSF NCAR CREDIT; ECMWF AIFS and FuXi via NVIDIA Earth2Studio; NeuralGCM inference in JAX), hyperparameter optimization, explainable AI (XAI), AI model evaluation, bias detection and correction

Physical and statistical modeling: Rapid Radiative Transfer Model, CESM Community Land Model v5.0, snow model SNICAR-ADv3, Bayesian inference and optimal-estimation retrieval

Geophysical data: DOE ARM Ka-band profiling radar observations, radiosondes, surface meteorology and broadband radiometers, satellite infrared radiances, ERA5 reanalysis, model simulations, cloud-hosted archives (Google GCP ARCO; AWS NOAA GOES)

Programming: Python, R, MATLAB, C, Fortran, Dask, xarray, NetCDF, Git/GitHub, Linux, conda/mamba environment management, HPC with Slurm (UMich Great Lakes, OU OSCER), LaTeX